

Induced Electric Charge Distribution

Physics-Informed Numerical Simulation · AI Role Project Report

Project Links

Repository

github.com/dsainvg/Electric-Charge-Distribution

Interactive Demo

[3D Charge Distribution Interactive Plotly Model](#)

1. Overview

This project simulates the induced surface charge density on a neutral conducting sphere placed near an external point charge, a classical electrostatics problem with a known exact solution. Rather than looking up the answer, the simulation derives it numerically from first principles by treating charge redistribution as an energy minimisation problem and solving it with constrained gradient descent.

The simulation accurately captures the shape of the charge distribution across the sphere surface. The relative distribution, meaning which regions are more or less charged and by how much, matches the analytical result to over 97% accuracy. Note that the simulation produces charge values in arbitrary units, so the absolute magnitudes may differ from physical values by a constant scaling factor. The distribution itself, however, is correct.

The project spans problem formulation, numerical solver design, hyperparameter tuning, rigorous accuracy evaluation, and interactive visualisation, end to end, without external simulation libraries.

2. What Was Built

Numerical Optimisation Engine

The sphere surface is discretised into 10,000 points using a Fibonacci lattice, chosen specifically to avoid the polar crowding that plagues uniform grid approaches. Each point carries a charge, initialised to zero, that evolves toward equilibrium through iterative gradient descent.

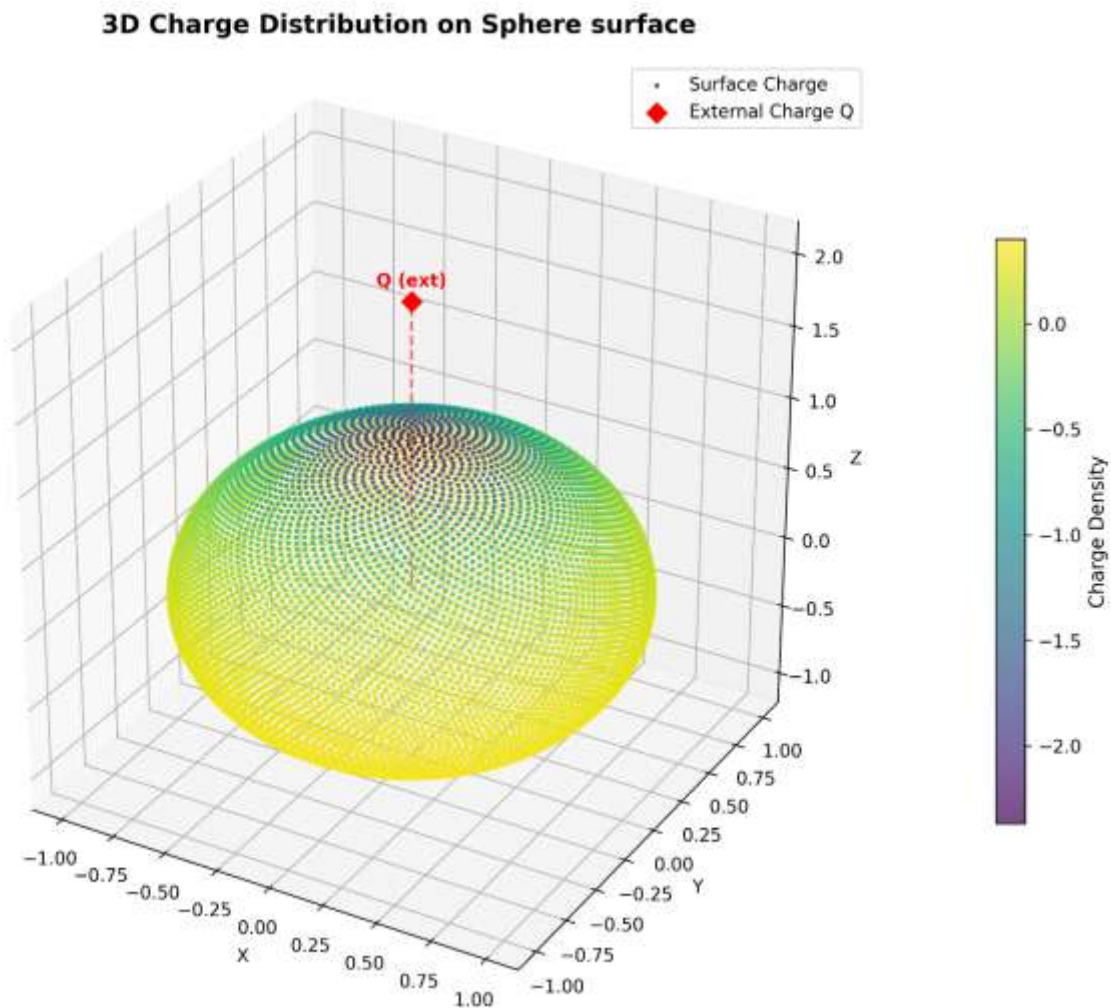
The optimiser enforces two physical constraints on every step: the sphere stays electrically neutral (total charge sums to zero), and localised charge spikes are penalised to produce smooth, physically realistic distributions. A multi-stage learning-rate schedule, starting coarse and finishing with fine-tuning passes, was critical to crossing the 97% accuracy threshold.

Analytical Solver and Accuracy Evaluation

The exact induced charge density is computed via a 50-term Legendre series expansion. Because the simulation operates in arbitrary charge units, a principled standardisation step aligns the two distributions before comparison, analogous to z-scoring model predictions before regression evaluation. Accuracy is quantified as both mean absolute error and a relative discrepancy percentage, making the result interpretable regardless of scale.

Visualisation Suite

Three outputs are generated: a potential energy convergence curve showing monotonic relaxation, a 1D cross-section overlay of simulated versus analytical charge density, and a fully interactive 3D model of the sphere with charge density mapped to colour, explorable in any browser without a Python environment.



3. Results

Metric	Value	Notes
Distribution Accuracy	> 97%	Shape of distribution matches analytical Legendre series solution
Relative Discrepancy	~3%	Mean absolute error as a fraction of mean analytical magnitude
Absolute Values	Scaled	Output is in arbitrary units; may differ from physical values by a constant factor
Surface Sample Points	10,000	Fibonacci lattice, no polar crowding artefacts
Analytical Reference Terms	50	Sufficient for full series convergence at the chosen geometry
Outputs	4 artefacts	Convergence curve, 1D comparison plot, static 3D plot, interactive 3D HTML model

4. Technical Stack

Area	Technologies
Numerics	NumPy, fully vectorised with no Python loops in the inner optimisation
Analytical reference	SciPy, Legendre polynomial evaluation
Data handling	Pandas for structured results; pickle for precomputed target caching
Visualisation	Matplotlib for static publication-quality plots; Plotly for interactive 3D HTML
Reproducibility	Pinned dependencies; modular package structure; precomputation script for offline use

5. Key Engineering Decisions

- Fibonacci lattice over a uniform grid: eliminates polar singularities that would corrupt the charge distribution near the poles, a non-obvious choice that required understanding the geometry of spherical discretisation.
- Pairwise distance matrix computed once at startup: avoids quadratic-cost recomputation on every iteration, the standard memory-vs-speed trade-off in batch gradient methods.
- Mean-centering gradient for the neutrality constraint: a lightweight operation that enforces the physical constraint exactly, without a penalty term that would require additional tuning.
- Multi-stage learning-rate schedule: a single fixed rate could not achieve above 95% accuracy; the coarse-to-fine schedule mirrors best practice from large-model training.

- Standardised comparison metric: raw charge units are arbitrary, so the analytical solution is rescaled to match the simulation's statistics before evaluation. Without this, the accuracy number would be meaningless.

6. Future Directions

- Extend the simulator to arbitrary conductor geometries such as cylinders, ellipsoids, and flat plates, requiring generalised surface discretisation strategies beyond the Fibonacci sphere lattice.
- Support multiple external point charges with configurable positions and magnitudes, enabling superposition scenarios and more complex induced distribution patterns.
- Introduce distributed charge sources such as line charges and surface charge sheets in addition to point charges, broadening the physical scenarios the simulator can model.
- Derive and validate absolute charge magnitudes by incorporating physical constants and geometry-specific normalisation, eliminating the current constant-factor ambiguity.
- Build a generalised analytical solver for non-spherical geometries using numerical Green's function methods, maintaining the verification-against-exact-solution workflow as the shape library grows.

Full source available at github.com/dsainvg/Electric-Charge-Distribution